

LEAD-SAFE

LeadSafe

Lead detection

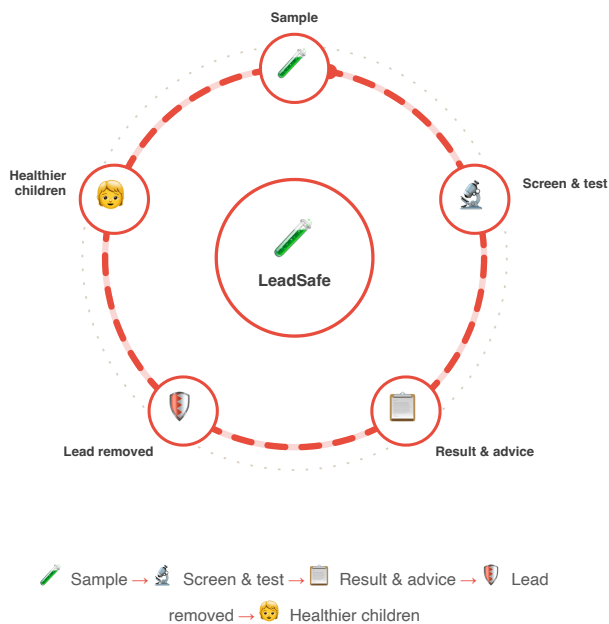
Bangladesh's mass childhood lead poisoning is detectable, costly, and demonstrably fixable

THE SITUATION IN BANGLADESH

In Bangladesh an estimated 35.5 million children carry blood lead levels at or above 5 µg/dL, the threshold for action, making the country the fourth most-affected in the world by number of children poisoned ¹. The damage is not abstract: lead exposure costs an estimated 20 million IQ points lost each year, valued at roughly USD 10.9 billion, on the order of 3.6% of GDP ¹. The poison hides in everyday life, flowing from more than 1,100 informal used-lead-acid-battery recycling sites, lead-adulterated turmeric and spices, contaminated cookware and paint ¹. Paint alone shows how porous enforcement is: despite a national 90 ppm limit set in 2018, a 2026 study found 42% of sampled paints exceeded it and 26.2% topped 1,000 ppm, with only 21.6% properly labeled ³, and the gaps cluster among smaller and informal producers where regulation reaches least ³.

OUR PITCH

LeadSafe does not sell tests; it sells trust — deploying local testing agents armed with handheld XRF analyzers, the same field technology the US Consumer Product Safety Commission relies on to screen consumer goods to within roughly 0.5% accuracy, to verify products and award a 'Lead-Safe' certification mark traders and brands can display ⁴. The wedge is proven: the Stanford-led turmeric intervention combined rapid lead detection with food-safety enforcement and education and drove adulterated market samples from 47% in 2019 to 0% in 2021, cutting tested blood lead levels a median 30% within 16 months ². That same playbook generalizes, because Pure Earth's screening across 25 low- and middle-income countries found lead in cookware, foodware, cosmetics, spices and toys — a broad and certifiable market beyond any single product ⁵. By making 'tested clean' a visible, sellable asset, LeadSafe aligns commercial incentive with the public-health goal rather than fighting it, closing the enforcement gap that leaves informal producers untouched ³. It is not a silver bullet — certification works only where buyers reward it — but in a market already stung by turmeric and paint scandals, demonstrated safety is becoming something worth paying for ¹.



SOURCES (5)

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